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Syllabus

CS577 Deep Learning

Instructor

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Yan Yan | Illinois Institute of Technology (iit.edu)

Course Description

An introduction to the field of deep learning, including neural networks, convolutional neural networks, recurrent neural networks, transformers, generative models, neural network compression and transfer learning. This course will benefit students' careers as a machine learning engineer or data scientist.

Course Outcomes

Upon successful completion of this course, you will be able to:

- Understand the foundation, major techniques, applications, and challenges of deep learning;
- Apply deep learning algorithms for solving real-world problems.

The learning outcomes will be assessed based on a combination of homework assignments and exams.

Course Materials

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Deep Learning, by Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016.

Software Requirements: Python, Pytorch or TensorFlow

Course Outline

The course consists of 8 modules that focus on the following key areas:

Module 1: Neural Networks

Key concepts

- Deep learning applications
- Introduction to deep learning
- Neural network
- Backpropagation - introduction and forward pass
- Backpropagation – backward pass

Readings

- Deep Learning, by Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016. Read Chapters Chapter 6.2, 6.4, 6.5

Module 2: Convolutional Neural Networks (CNNs)

Key concepts

- CNN for images
- Convolution layers
- Pooling and other layers
- CNN applications

Readings

- ImageNet Classification with Deep Convolutional Neural Networks, Alex Krizhevsky, Ilya Sutskever, Geoffrey E. Hinton, NeurIPS 2012
- Deep Learning, by Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016. Read Chapters 9.1, 9.3

Module 3: Recurrent Neural Networks (RNNs)

Key concepts

- Introduction to RNN
- Long Short-term Memory (LSTM)
- Learning on RNN

Readings

- Fundamentals of Recurrent Neural Network (RNN) and Long Short-Term Memory (LSTM) Network, Alex Sherstinsky.
- Deep Learning, by Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016. Read Chapters 10.2, 10.3, 10.5

Module 4: Deep Learning Tips

Key concepts

- Introduction
- ReLU and Maxout
- RMSProp
- Early stopping and Regularization
- Dropout

Readings

- Maxout Networks, Ian J. Goodfellow, David Warde-Farley, Mehdi Mirza, Aaron Courville, Yoshua Bengio; ICML 2013
- An overview of gradient descent optimization algorithms, Sebastian Ruder;
- Dropout: A Simple Way to Prevent Neural Networks from Overfitting, Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, Ruslan Salakhutdinov; Journal of Machine Learning Research 15(56):1929−1958, 2014.
- Deep Learning, by Ian Goodfellow, Yoshua Bengio and Aaron Courville, 2016. Review Chapters 6.2, 6.4, 6.5

Module 5: Transformers

Key concepts

- Self-attentions
- Seq-2-Seq
- Transformer Encoder
- Transformer Decoder
- Transformer Training

Readings

- Attention Is All You Need, Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, Illia Polosukhin; NeurIPS 2017

Module 6: Generative Models

Key concepts

- Introduction to Generative Models
- Generative Adversarial Networks (GANs)
- Diffusion Models (DMs)

Readings

- Generative Adversarial Networks, Ian J. Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, Yoshua Bengio; NeurIPS 2014
- Denoising Diffusion Probabilistic Models, Jonathan Ho, Ajay Jain, Pieter Abbeel; NeurIPS 2020

Module 7: Neural Network Compression

Key concepts

- Introduction to efficient AI
- Network Pruning
- Knowledge Distillation
- Quantization

Readings

- An Overview of Neural Network Compression, James O' Neill

Module 8: Transfer Learning

Key concepts

- Introduction to transfer learning
- Fine-tuning
- Multitask Learning
- Domain adversarial training
- Zero-shot learning

Readings

- Transfer Learning Guide: A Practical Tutorial With Examples for Images and Text in Keras
- Domain-Adversarial Training of Neural Networks, Yaroslav Ganin, Evgeniya Ustinova, Hana Ajakan, Pascal Germain, Hugo Larochelle, François Laviolette, Mario Marchand, Victor Lempitsky; Journal of Machine Learning Research, 2016
- Zero-Shot Learning - A Comprehensive Evaluation of the Good, the Bad and the Ugly, Yongqin Xian, Christoph H. Lampert, Bernt Schiele, Zeynep Akata; TPAMI 2018

Course Structure and Learning Activities

There are 8 content modules in this course and each module may take about 1 hour to complete. The final module consists of your final exam for the course.

This course is comprised of the following elements:

Learning Activities:

- **Readings**: Each module may include several required and/or supplemental readings.
- **Video Lessons**: In each module, the concepts you need to know will be presented through a collection of short videos. You may stream these videos for playback within the browser by clicking on their titles.
- **Practice Quizzes**: Each module will include some practice quizzes, intended for you to assess your understanding of the topics. You will be allowed unlimited attempts at each practice quiz. There is no time limit on how long you take to complete each attempt at the quiz. These quizzes do not contribute toward your final score in the class.

Assessments:

- **Discussion Forum**: This course has a place for you to interact with other learners about class-related topics.
- **Summative Module Assessments:** Each module will include at least one summative module assessment. You will be allowed one attempt every eight hours for each assessment. There is no time limit on how long you take to complete each attempt at the assessment. Your highest grade will be recorded.
- **Final Assessment:** This course will contain one summative course assessment. You will be allowed one attempt for the assessment. Before taking the exam, please make sure you are in a place with reliable internet connection. No retakes will be granted for the lack of internet access. You are in an online program and the use of the Internet is a requirement.

How to Pass This Course

Guidelines for completing and submitting each assigned course activity is posted along with the assignment. Assignments can be submitted at any time as you move through the module. *No late assignments will be accepted.* In case of extenuating circumstances beyond your control that prevent the submission of an assignment or exam, you have to enter a request with the program advisor and the instructor.

To qualify for a Course Certificate, simply start verifying your coursework at the beginning of the course and pay the fee. Coursera Financial Aid is available to offset the registration cost for learners with demonstrated economic needs. If you have questions about Course Certificates, please see the help topics here.

Also note that this course is part of the Master of Data Science program offered by Illinois Institute of Technology. By earning a Course Certificate in this course, you are on your way toward earning a Specialization Certificate in this topic. See more information about the program here.

If you choose not to pay the fee, you can still audit the course. You will still be able to view all videos, submit practice quizzes, and view required assessments. Auditing does not include the option to submit required assessments. As such, you will not be able to earn a grade or a Course Certificate.

The following table explains the breakdown for what is required in order to pass the class and qualify for a Course Certificate. You must pass each and every required activity in order to pass this course.

Activity	Required?	Number per Course	Estimated Time per Module	% Required to Pass	% of Total Grade
Lecture Videos	Yes	3-6 per module	.5-1 hour	N/A	N/A
Practice Quizzes	No	3-6 per module	.5 hour	N/A	N/A
Discussions	No	1 per course	1 hour	N/A	N/A
Summative Module Assessments	Yes	1 per module	.5 hour	80%	7.5%/each module (60%)
Summative Course Assessment	Yes	1-2 per course	1-3 hours	80%	40%

Letter Grades

Letter grades are used for the final grade. Information about IIT grading system can be found in the Graduate Student Handbook.

Letter Grade	Description	Points	Percentage
A	Excellent	4.00	90-100
B	Above Average	3.00	80-89.99
C	Average	2.00	70-79.99
E	Fail	0.00	Under 70%

Getting and Giving Help

- Use the Learner Help Center to find information regarding specific technical problems. For example, technical problems would include error messages, difficulty submitting assignments, or problems with video playback. If you cannot find an answer in the documentation, you can also report your problem to the Coursera staff by clicking on the *Contact Us!* link available on each topic's page within the Learner Help Center.
- Use the flag icon under each item to report errors in lecture video content, assignment questions and answers, assignment grading, text and links on course pages, or the content of other course materials.
- Familiarize yourself with Coursera's policy on Accessibility.

Academic Integrity

Your attentiveness to academic integrity reflects the value you place on your own work and the work of others. In addition to Coursera's Honor Code, we also have high expectations for conduct during course participation.

Discussion Forums: Expectations

Sharing an online course with other avid learners like you gives you a unique opportunity to share, collaborate, and learn from others and their experiences, and helps you reinforce your understanding of the topics of the course. Interacting in the Discussion Forums is a great way to engage with your online community. We know that it is not possible to read every discussion forum post, so we recommend that you read those that interest you; and reply when you can contribute. The forum is part of your class activities and everybody is expected to act professionally and be civil and respectful of others in your class. Failure to meet these expectations may be considered a break in the Academic Code of Conduct and may result in your removal from the course. Please, check tips and helpful tools to interact in discussion forums in this document

Academic Code of Conduct

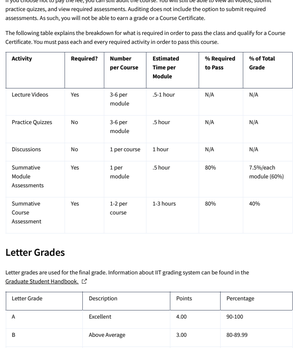
Above all else, learners are expected to ensure that their conduct helps to create an atmosphere conducive to learning and the interchange of knowledge. While it is understood that some of these items are subject to interpretation, learners should nonetheless endeavor to:

- Be respectful of fellow learners.
- Not discriminate against fellow learners in any manner.
- Conduct peer reviews in a timely manner and give useful feedback on what was done well, helpful suggestions for how to improve, and specific comments about why you gave the grade you chose to assist peers in their learning.
- Turn assignments in on time and follow instructions on all assignments including those affecting the use of technology.
- Be truthful in all communication, which includes, but is not limited to, avoiding academic dishonesty.

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SYLLABUS

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Instructor

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